

Shounak Ghosh

shounakghosh42@gmail.com | sh42.me/pf | github.com/shahank42

Experience

Software Development Engineer, Heizen – Hyd, India

Aug 2026 – Present

Promoted from Software Development Engineer Intern

Jul 2025 – Aug 2026

- **Heizen Studio** (AI-native product management and SDLC platform)
 - Built a Google Meet bot that joins calls, records, transcribes and summarizes meetings, and hardened it against bot detection, raising successful meeting joins from 10% to 60%
 - Reduced CPU by 17% and GPU by 50%+ in the Electron-based Heizen Meets app by eliminating redundant re-renders and optimizing animations; shipped on-device STT model transcription alongside cloud providers
 - Built LLM agent infrastructure used across Heizen Studio and Helix the AI agent: platform-integrated tool calling, browser automation, document ingestion into LLM-backed knowledge bases, and context management for 300k+ token workloads used by over 10+ teams
- **Internal tools** used company-wide
 - Built **Atlas**, an interactive company knowledge graph (ReactFlow) of domain-specific knowledge, adopted by employees throughout the company
 - Built **Accounts**, a finance tool that reconciles bank statements and RazorpayX ledgers into MIS sheets and AP/AR reports, saving the accounting team 16 hrs/month
- **PsyFi**, a personal finance and investment tracker (shipped to production, primary engineer)
 - Built most of the web app and its backend: XIRR, capital-gains and volatility engines, plus multi-tenant Plaid integration
 - Provisioned production environments on AWS via ECS with automated CI/CD and unit tests; contributed to SOC2 technical controls
 - Cut DB query time by up to 92% on financial transaction mapping queries and improved Next.js load times with React Server Components; added SEO, taking bounce rate from 40% to 20%
- **Jetfuel**, an AI website builder for construction companies
 - Built an AI website editor (v0/Lovable-style) on Sandpack with automatic error fixing, and integrated Google Ads and Umami Analytics so sites ship monetizable out of the box

Backend Intern, Aarish Technologies Inc. – QC, Canada (Remote)

October 2024 – March 2025

- Architected high-performance H.264 video streaming and archival pipeline for military mesh network application, optimizing latency and bandwidth using FastAPI, ffmpeg, and GStreamer
- Designed fault-tolerant two-way communication protocol for mesh networks by developing custom UDP packet encoding for ESP32 nodes in unstable network conditions
- Devised a setup for audio classification by creating a full-stack web application with React and FastAPI, allowing integration with TensorFlow; the solution is currently deployed for defence-sector clients

Projects

SophistAI – Your Personal Syllabus Navigator

sophistai.app

- Conceptualized and developed an AI-powered mind-map generator that automatically sorts user-submitted syllabi into hierarchical study pathways; this reduced student research time by 6 hours per week
- Crafted an intuitive and responsive mind map; backed by a robust backend that queries a hosted PostgreSQL database
- Launched as a student-focused SaaS: 10k+ site views in the first week, 700+ registered users from various colleges in India (with some from USA and Japan), plus end-to-end payments, SEO optimization, and real-time analytics

Achievements

- Placed at 3 hackathons: **1st** at Diversion 2k25 (200+ teams, SophistAI), **1st runner-up** at Statuscode1 (150+ teams, NF-Tree), and **National Finalist** at Smart India Hackathon 2023
- Secured a **top 1% national ranking** in Mathematics in the AISSCE and AISSE

Education

Kalyani Government Engineering College, Nadia – BTech in Computer Science and Engineering

July 2026

Skills

Languages: TypeScript, Python

Fullstack: React, Next.js, Svelte, Svelte-kit, Electron, Node.js, NestJS, Bun, Hono.js, Express, Prisma, DrizzleORM, Jest, Vitest

Infrastructure: AWS, PostgreSQL, Redis, BullMQ, Grafana, SQLite, Cloudflare Workers, Docker, Convex

AI: LLMs, Context Engineering, AI SDK, Mastra, Langfuse, VectorDB, GraphDB, MCP